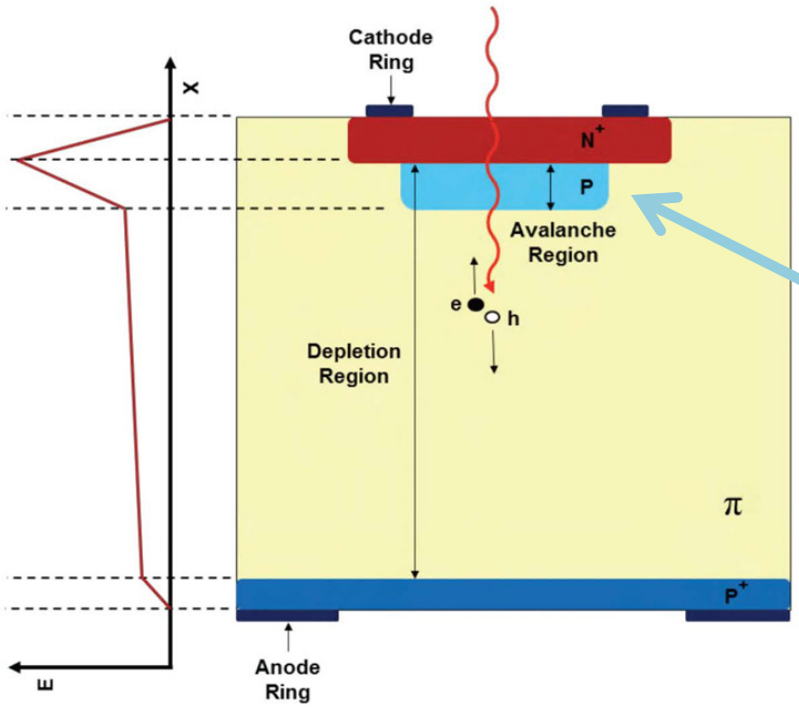
Time Resolution of Ultra-Fast Silicon Detectors (thin LGAD)

Hartmut F-W Sadrozinski
SCIPP, UC Santa Cruz

- Time resolution wrt
  - sensor thickness,
  - doping profile,
  - neutron fluence,
  - temperature
- First signals out of AC-LGAD

Low-Gain Avalanche Detectors (LGAD)



LGAD were introduced by CNM Barcelona.

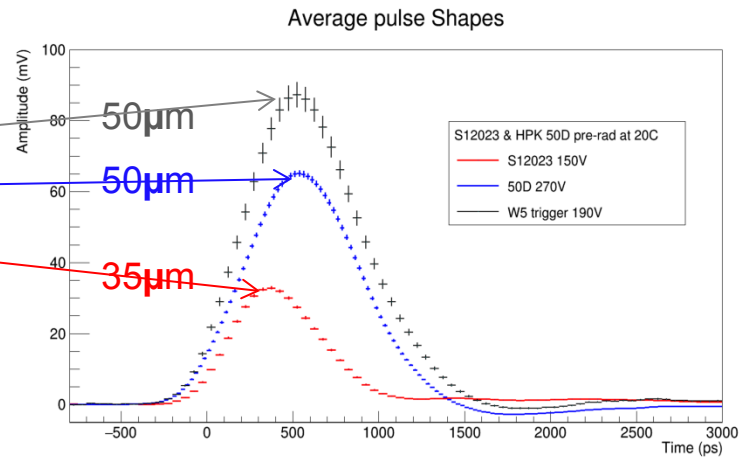
Principle:

Add to n-on-p Silicon sensor an extra thin p-layer below the junction which increases the E-field so that charge multiplication with **moderate gain** of 10-50 occurs without breakdown.

Timing characteristics depend on both the bulk (i.e. thickness) and the multiplcation layer.

Manufacturers of LGAD (30 µm – 300 µm):
CNM Barcelona: Mar
HPK Hamamatsu
FBK Trento: Valentina

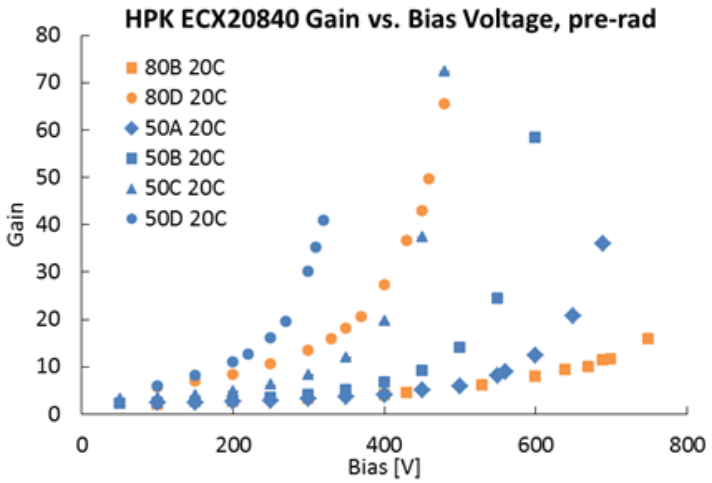
very similar behavior with exception of breakdown voltage and special design features .



HPK LGAD: 80 µm -> 35 µm

50 & 80 µm thick LGAD
(HPK 50D & 80D)
capacitance 2.7 pF
For HGTD expect C(50 µm) =3.4 pF

Neutron fluences up to 6e15 n/cm²



35 µm thick LGAD
(HPK sample B35)
capacitance 4.6 pF
For HGTD expect C(35 µm)= 4.8 pF

Higher Doping Concentration
-> Higher field, lower bias
Thicker sensors:
->Larger depletion voltage, higher bias

Comparison of 50 µm & 35 µm at 3 fluences pre-rad, 5e14, 1e15 n/cm²

Measured in the UCSC β-telescope pre-rad and post-rad at -20C & -30C

Parametrization of Time Resolution σ_t

$$\sigma_t^2 = \sigma_{TimeWalk}^2 + \sigma_{LandauNoise}^2 + \sigma_{Distortion}^2 + \sigma_{Jitter}^2 + \sigma_{TDC}^2$$

$$\sigma_{Jitter} = \frac{N}{dV/dt} \approx \frac{t_{rise}}{S/N}$$

$$\sigma_{TimeWalk} = \left[t_{rise} * \frac{V_{th}}{S}\right]_{RMS} \propto \left[\frac{N}{\frac{dV}{dt}}\right]_{RMS}$$

$\sigma_{LandauNoise}^2$

$\sigma_{Distortion}^2$

Signal slope dV/dt rules:

-> Fast signal: thin sensors

-> Large signals: gain

-> Fast electronics

Time walk is corrected with CFD method

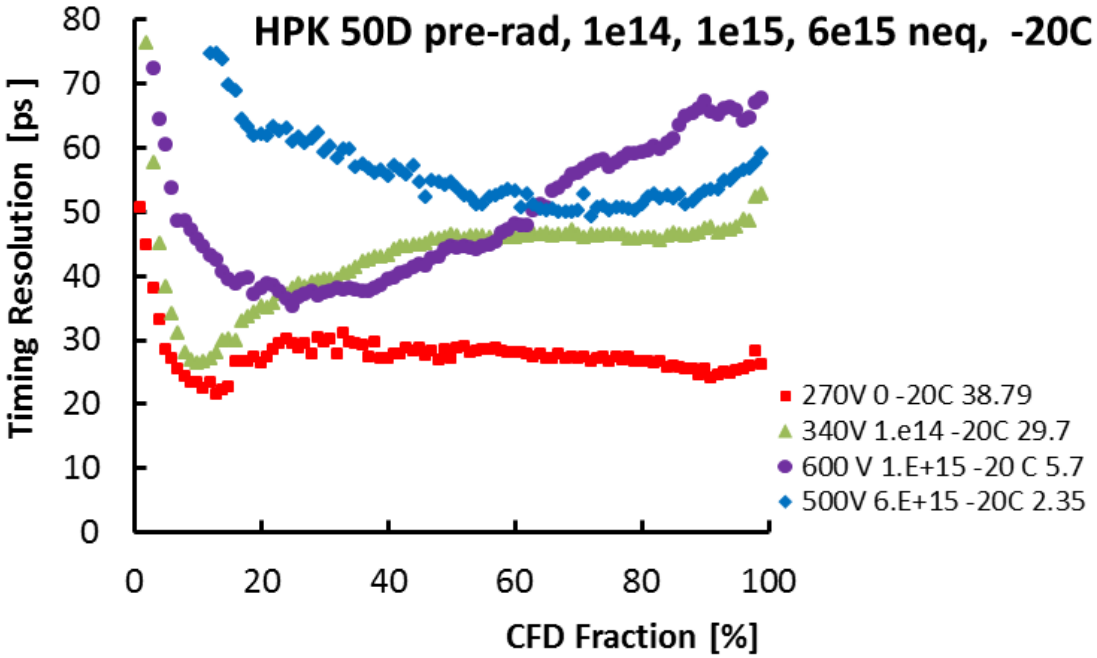
…. and need low noise!

Signal fluctuation due to non-uniform charge deposition Is often the ultimate limit, is minimized in thin sensors

Need uniform operating / weighting field

Time Stamp by CFD Method

The Constant-Fraction Discriminator (CFD) method assigns the time at which the pulse crossed a certain fraction of the pulse height Pmax.
The optimal CFD percentage is determined by a scan of the threshold..



Scan of the CFD threshold for 3 representative fluences..

The optimal CFD threshold to reach the best time resolution is at 10-20 % at fluences below 1e15 n/cm^2 and at > 50% above 1e15 n/cm^2.

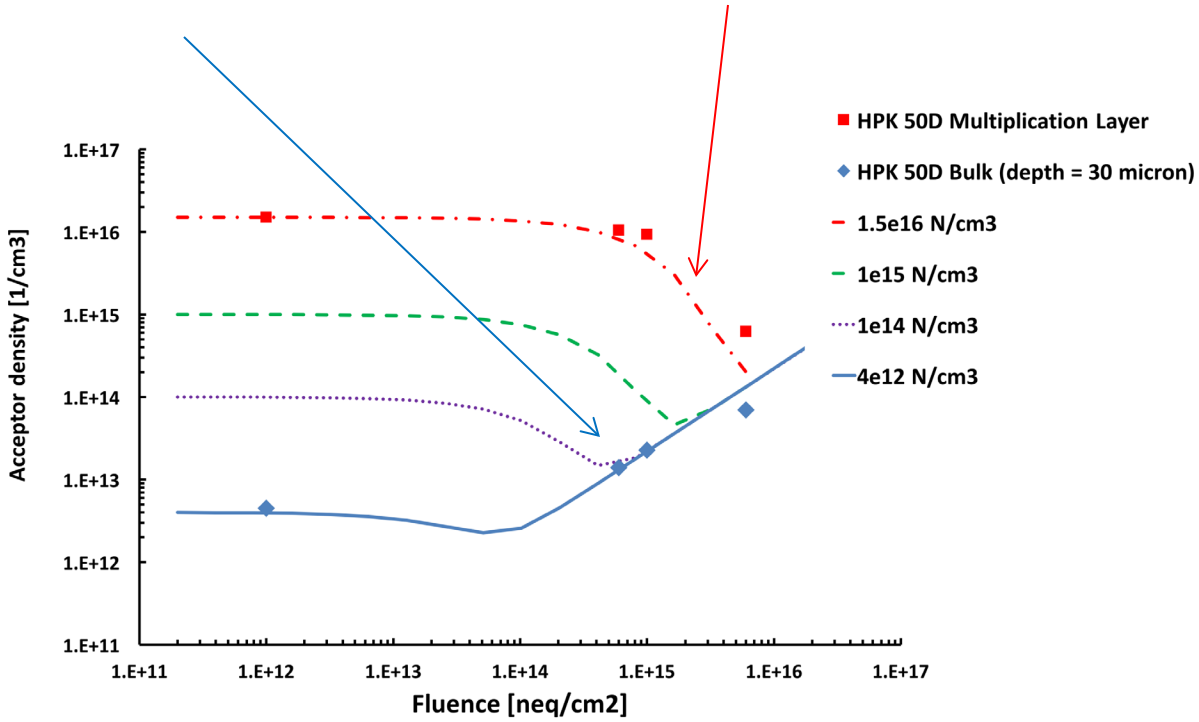
The reason is a change in pulse shape introduced by the acceptor removal process in the multiplication layer (- see back up)

Z. Galloway et al, arXiv:1707.04961

Radiation Hardness of LGAD: Acceptor Removal

An LGAD specific effect is the gain decrease due to acceptor removal in the gain layer.

Acceptor density vs. fluence:
Increases in low density bulk, decrease in high density gain layer.



Z. Galloway et al, arXiv:1707.04961

Data from C-V scans
Simulation from WF2 incorporating G. Kramberger's work

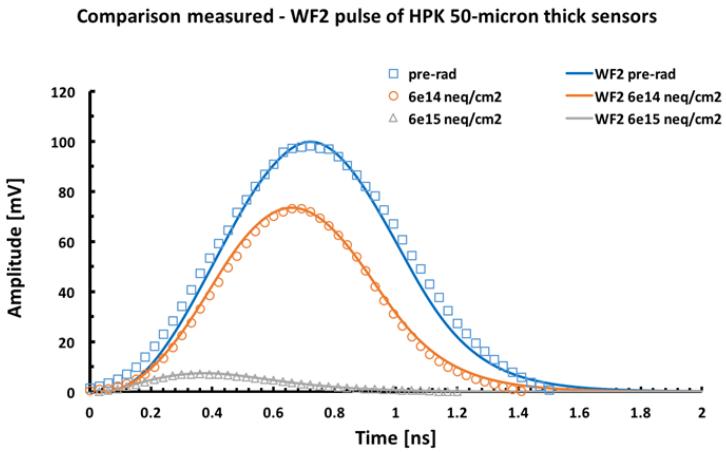
Technology Mitigation:
see following talks by Valentina, Mar et al.

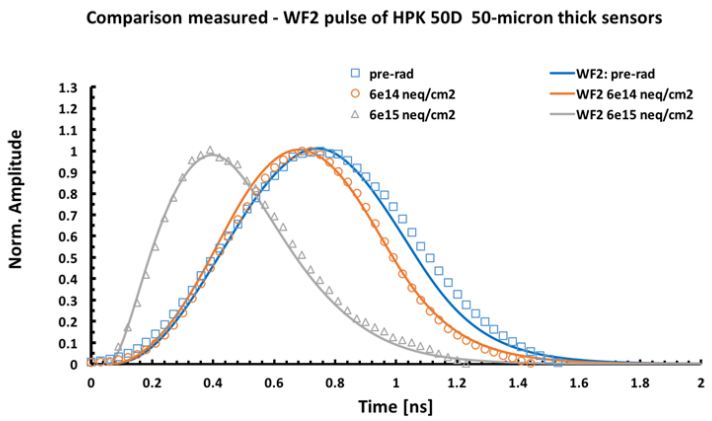
Radiation Effect of LGAD

Small thickness permits high voltage biasing ->
part of gain in the bulk, smaller rise time.

Radiation campaigns with CNM, HPK and FBK LGAD by RD50, ATLAS and CMS
Here report on 1mm HPK 50D (i.e. 50μm), data taken at -20 °C, and -30 °C.
Neutron fluence steps: **0**, 1e14, 3e14, **6e14**, 1e15, 2e15, 3e15 (HGTD), **6e15** n/cm^2

Decrease of gain **but also** **Decrease of rise time**

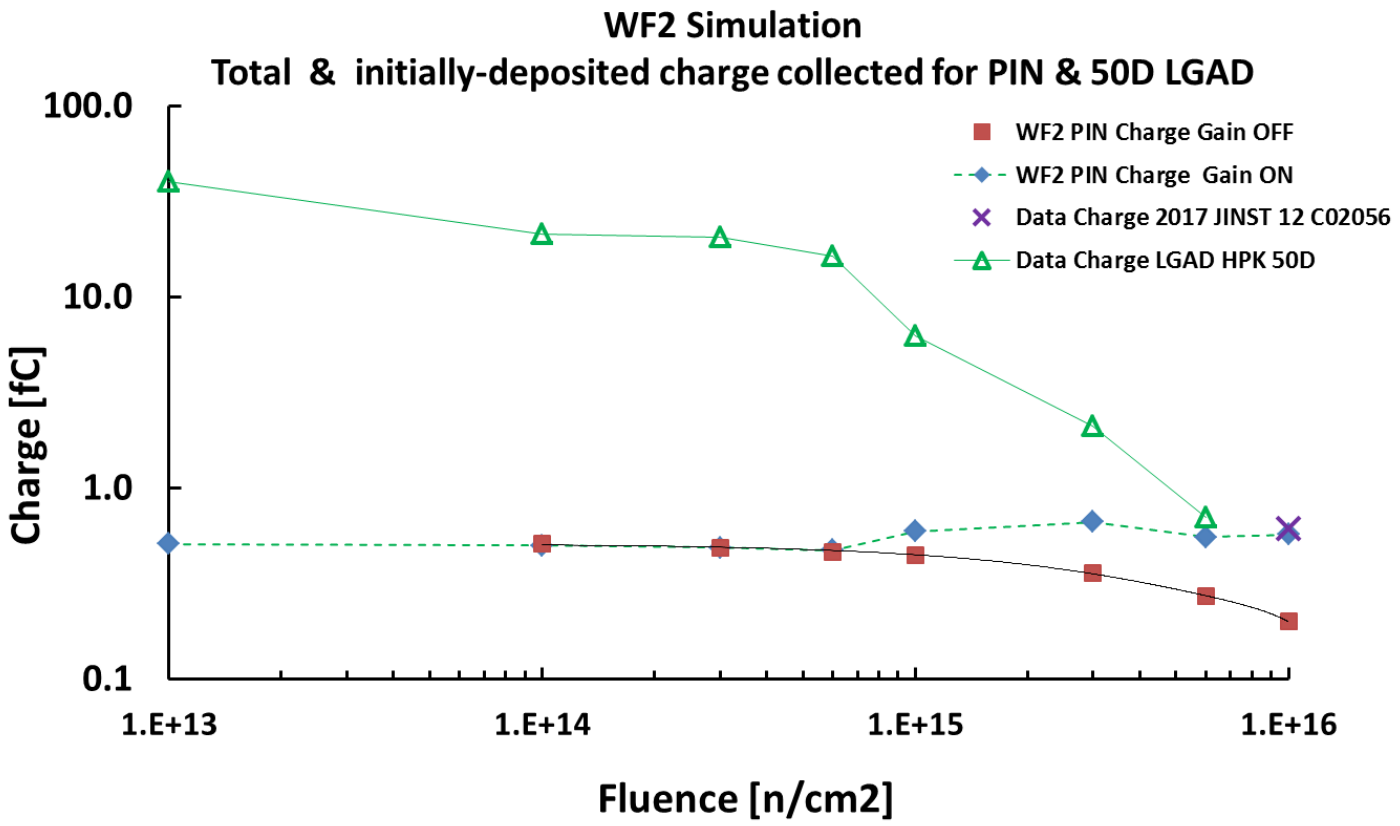




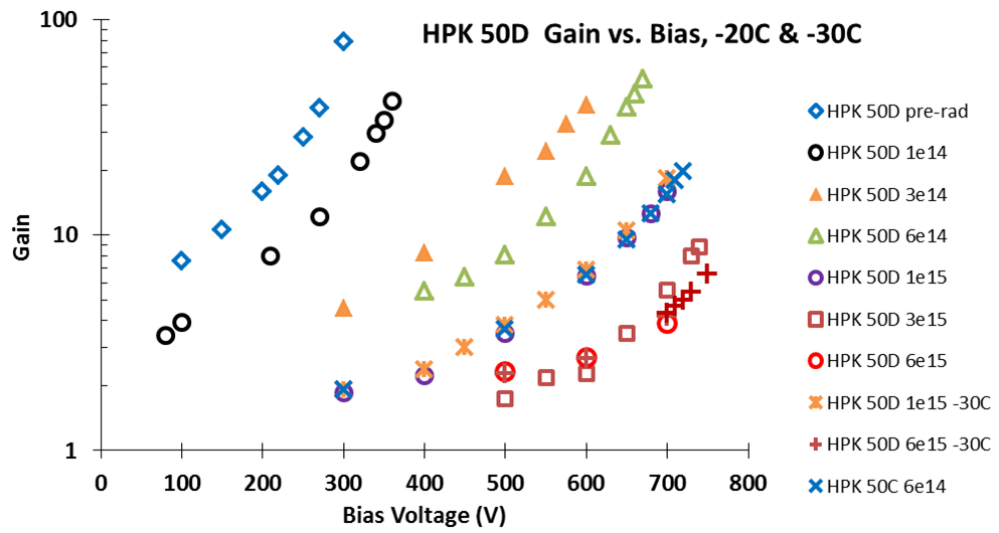
Z. Galloway et al, arXiv:1707.04961

How to calculate the Gain

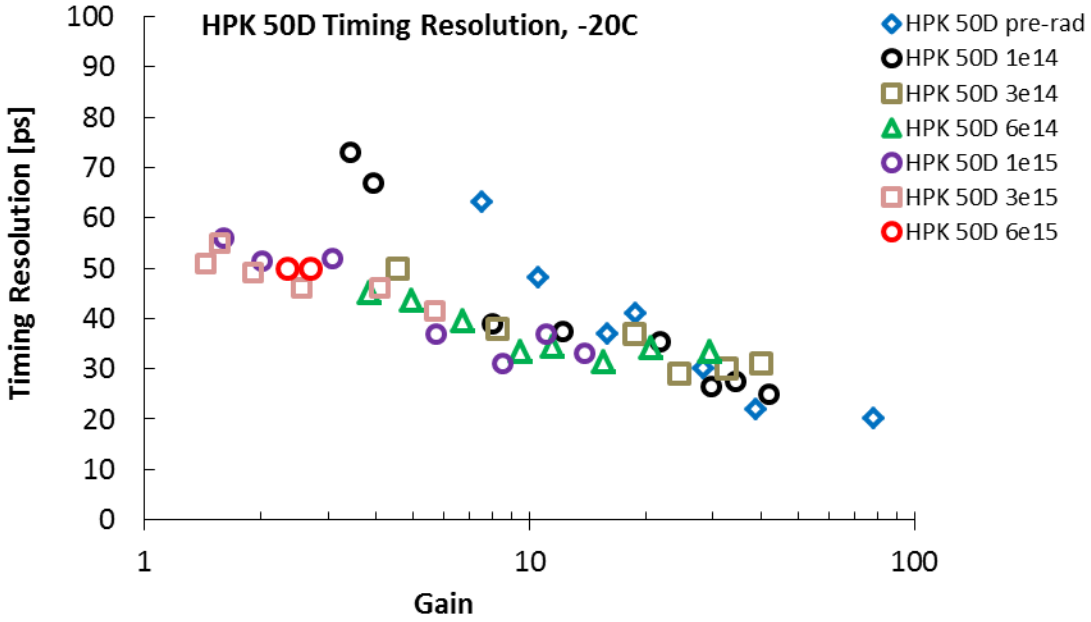
To characterize the multiplication process we define the gain as
Gain = collected charge / original deposited charge collected for a PIN (i.e. no gain)
The original PIN charge is calculated with Weightfield WF2.
The WF2 simulation is validated by the correct prediction of the collected charge of a real 50 µm PIN measured by E. Currás et al, (2017_J._Inst._12_C02056)



Gain & Time Resolution for 50µm HPK LGAD



After irradiation, the gain loss can be restored by increasing the bias.
That works at fluences below 1e15 n/cm^2
For higher fluences the detector breakdown limits the bias and the maximum gain is reduced.

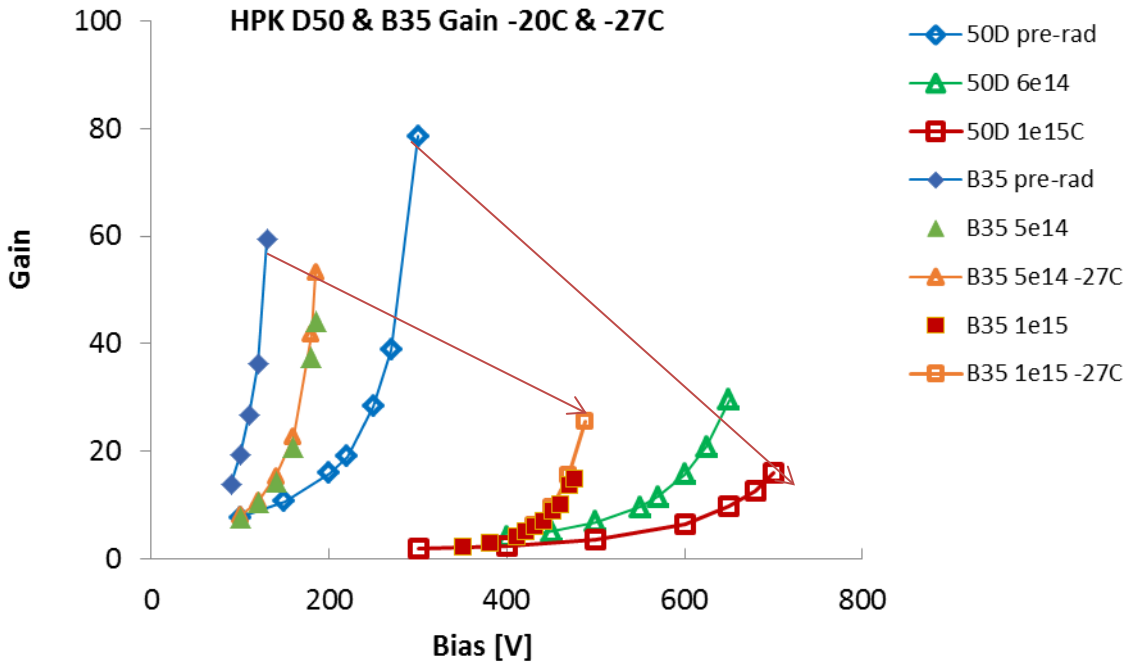


As expected the time resolution improves with increased gain.
At low fluences the low bias creates a low drift field which reduces the drift velocity.
At high fluences increased noise leads to flattening of the time resolution vs. gain curve.

Z. Galloway et al, arXiv:1707.04961

SCIPP
Hartmut F.-W. Sadrozinski, "Time Resolution Of UFSD", HSTD1
INFN

Bias Dependence of Gain: 35 µm vs. 50µm



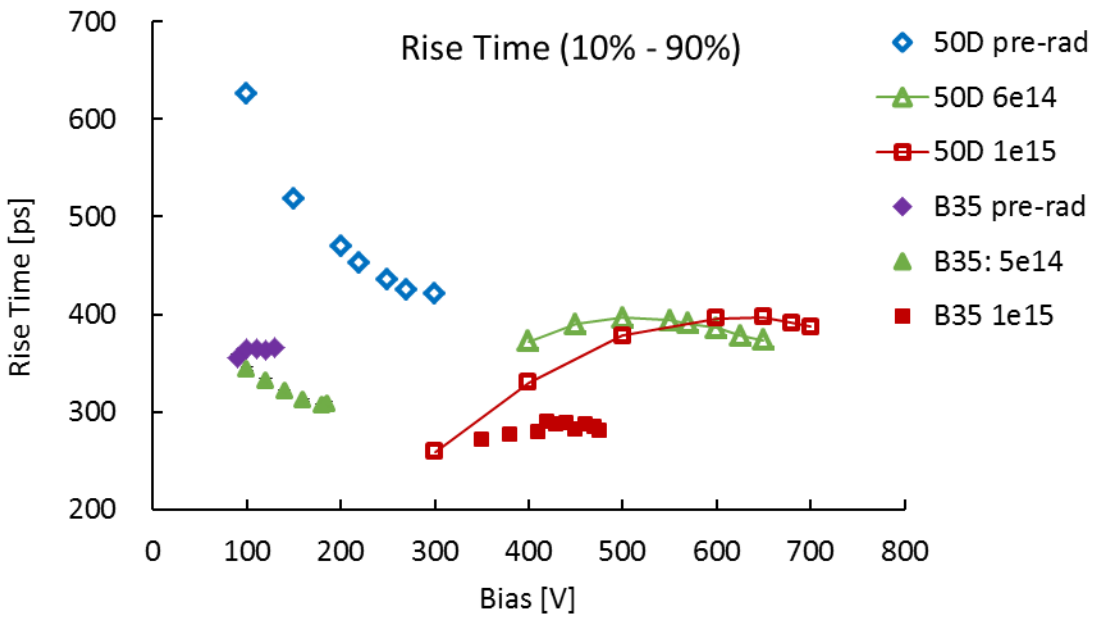
50 µm:
Bias increase large between pre-rad and 6e14 n/cm^2
and small between 6e14 n/cm^2 and 1e15 n/cm^2

35 µm:
Bias increase small between pre-rad and 5e14 n/cm^2
and large between 5e14 n/cm^2 and 1e15 n/cm^2

Lower bias voltage for 35 µm for same gain.

Bias voltage gap of ~ 200V between 35 µm and 50 µm is approximately preserved through radiation.

Rise Time (10%-90%) 35µm vs. Bias -20C



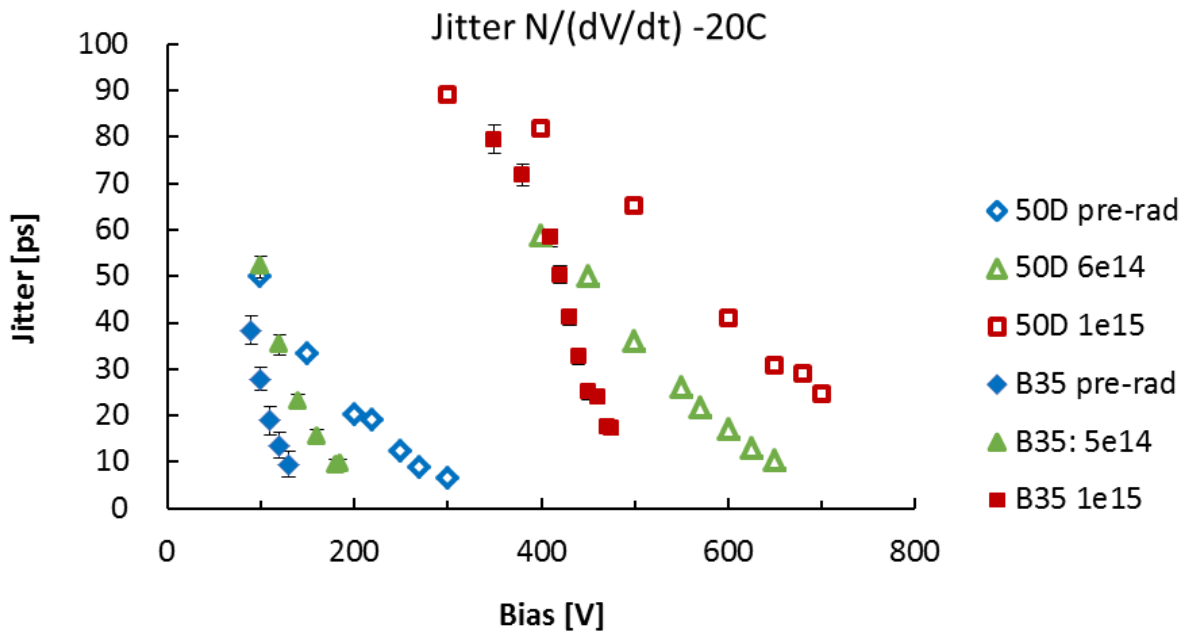
As expected the rise time for **35 µm** is significantly **lower** than for 50 µm.

Pre-rad:
low drift field evident.

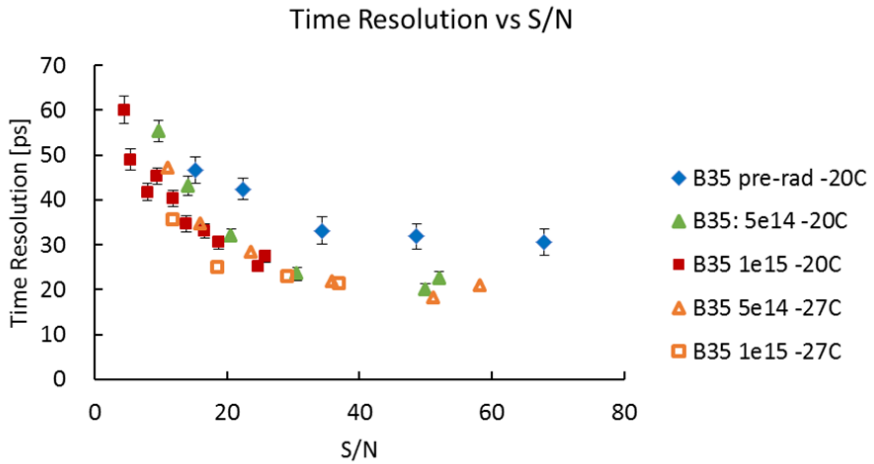
At larger fluence the Increase at high bias is due to the delayed holes from the gain process.

-> Bias voltage dependence of time resolution is a trade-off between two effects of increasing gain: increased signal and increased delayed hole signal.

Jitter: 35 µm vs. 50 µm



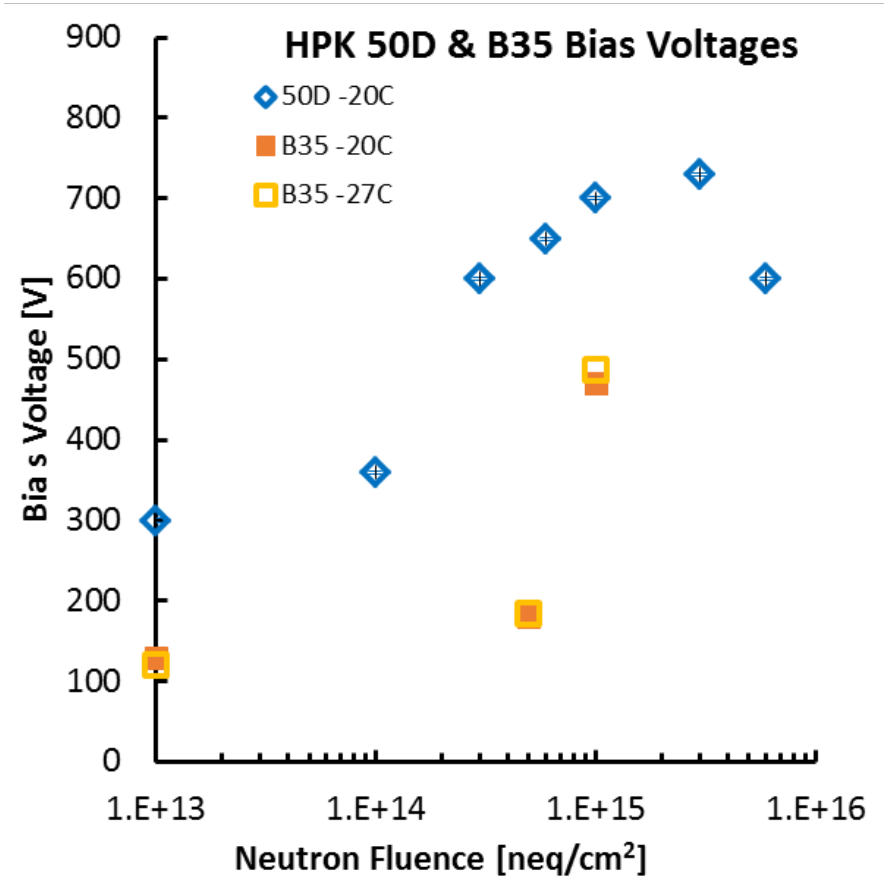
Jitter ≈ (rise time)/(S/N) is lower for 35 µm than for 50 µm because of the lower rise time

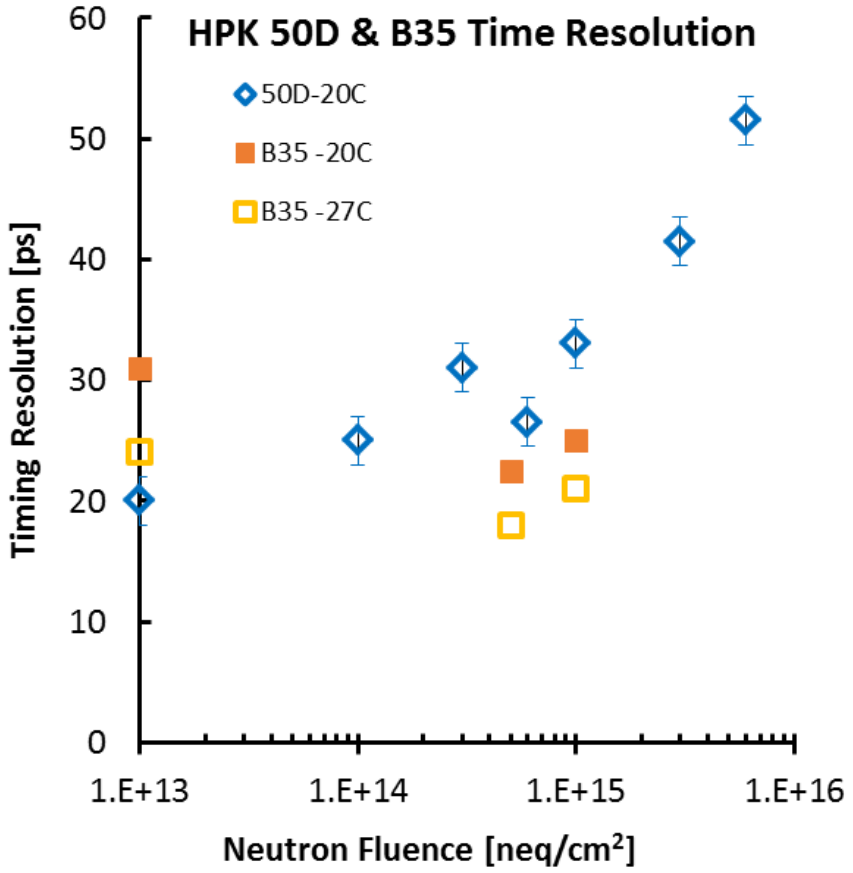


Jitter becomes important when it is as large as the Landau fluctuation of ~ 20 ps
i.e. above fluence of 1e15 n/cm^2

Score Card 50 µm vs. 35 µm

For every fluence the operating bias is selected as the bias where the time resolution is optimized.

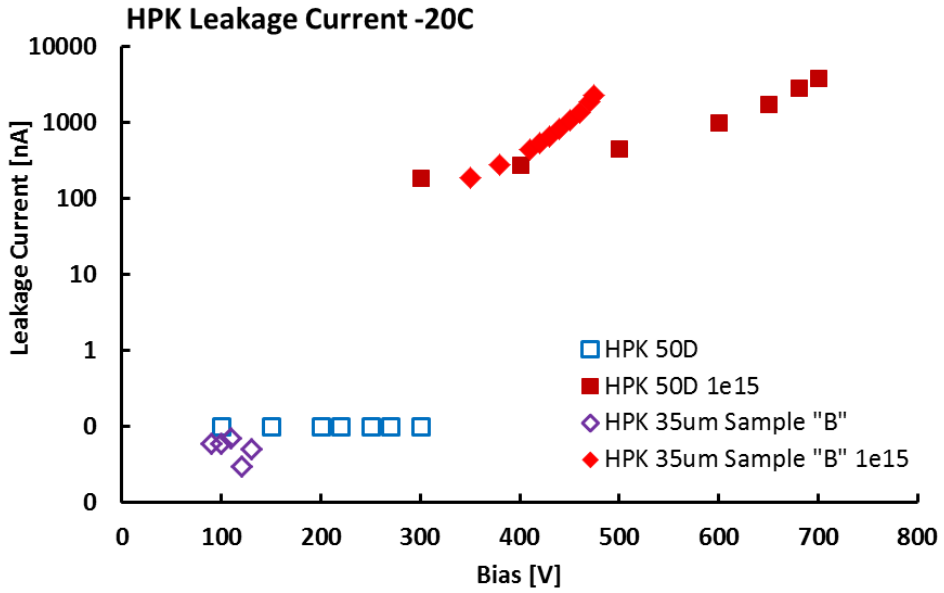


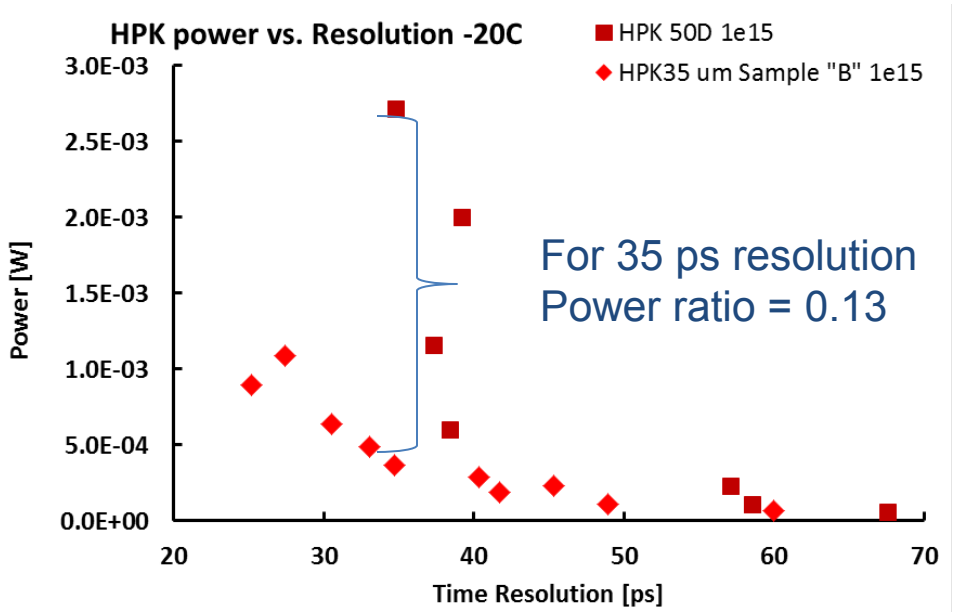


HPK "B" 35µm:
has lower bias voltage,
AND large temperature effect

AND better time resolution

Leakage Current & Power vs. Timing resolution





Power Consumption:

B35 wrt 50D has
lower current,
lower bias
-> lower power

Comparison of power for same time resolution

Understanding of Time Resolution

$$\sigma_t^2 = \sigma_{TimeWalk}^2 + \sigma_{LandauNoise}^2 + \sigma_{Distortion}^2 + \sigma_{Jitter}^2 + \sigma_{TDC}^2$$

$\sigma_{Jitter} = \frac{t_{rise}}{S/N}$ OK

$\sigma_{TimeWalk} = t_{rise} * \frac{V_{th}}{S}]_{RMS} \propto \left[\frac{N}{\frac{dV}{dt}}\right]_{RMS}$ OK

Go thin, have large bias voltage reach

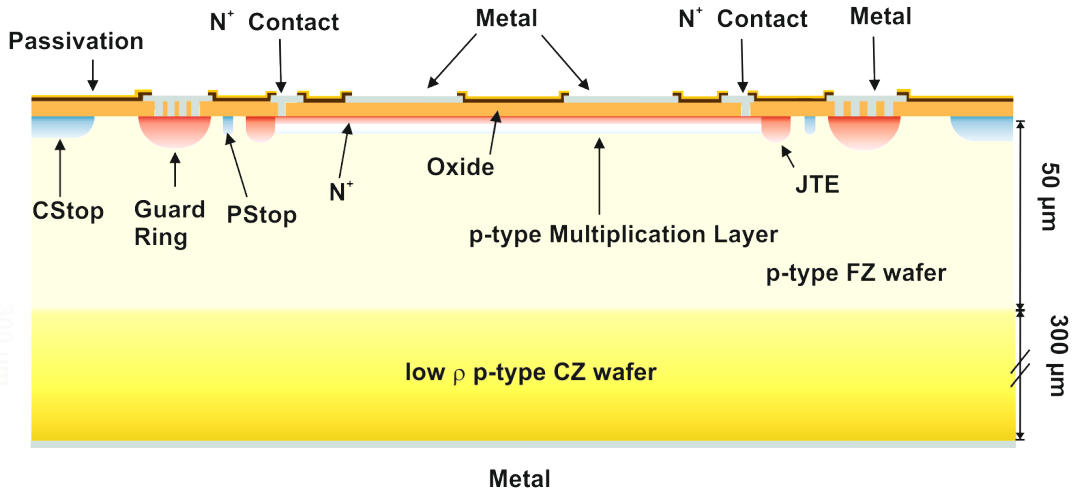
$\sigma_{LandauNoise}^2$ OK

Go thin, use lowest CFD threshold

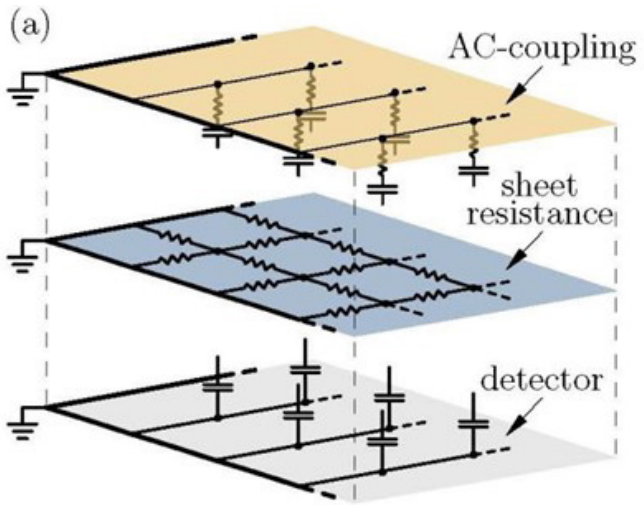
$\sigma_{Distortion}^2$

Develop AC-LGAD

AC-LGAD



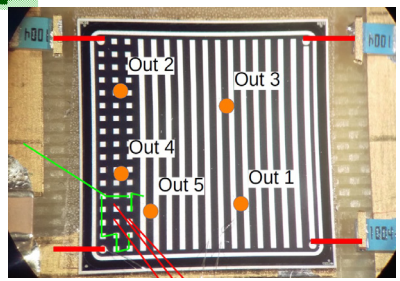
Principle:
Metal contacts are the only segmented layer:
Unsegmented sheets:
- Coupling oxide
- N^+ electrode
- p-multiplication layer



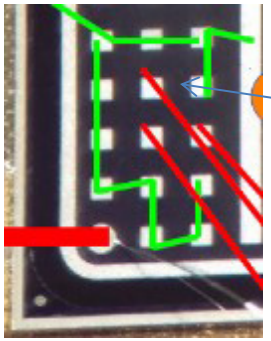
SPICE simulation of circuit
Optimization of
- sheet resistance of N-layer
- doping profile of p-multiplication layer
- Oxide thickness of coupling layer

Recording of pulse shapes in pixels / strips by
IR laser
α-particles
B-particles
Good agreement between SPICE and scope pulses!

First Signals from AC-LGAD

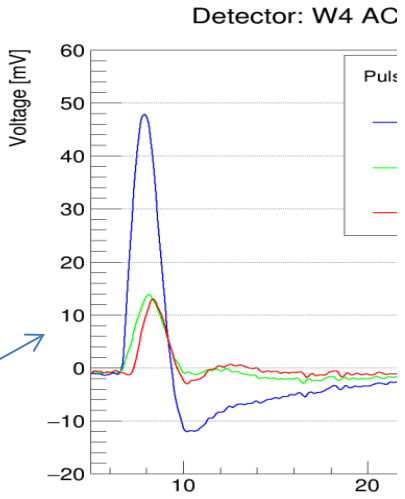


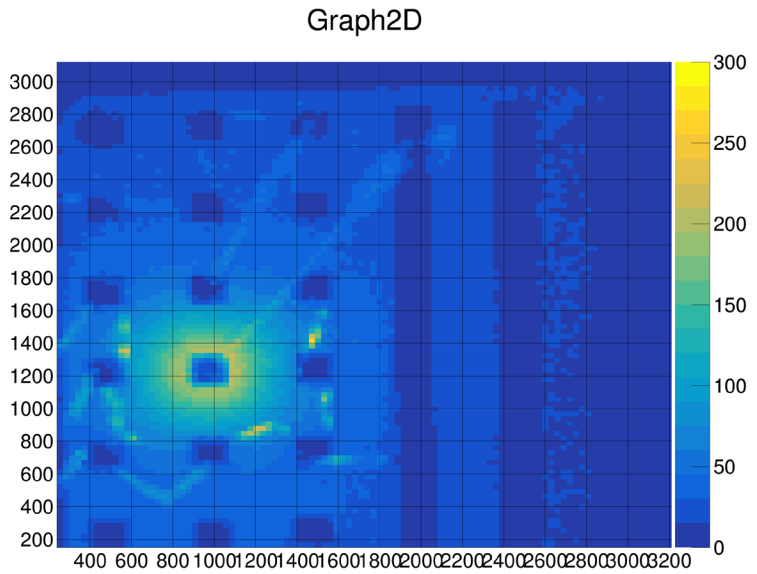
AC-LGAD with Pixels on 300µm pitch were produced as part of a HGTD prototype run at CNM.
No optimization of the sheet resistance!



Bias reach of AC-LGAD larger than corresponding single-pad LGAD.

Fast signals from laser showing pulse height difference between neighboring pixels





Sheet resistance of N+-layer not optimized: yet signal spread only a few 100 µm in a laser scan.

Localization shown to work by triangulation with signal pulse height

-> RD50 Common Project

Summary

Measure in β-telescope two LGAD from HPK with thickness 35 µm and 50 µm.

- Pre-rad both sensors have similar performance
  30 ps (35 µm) vs. 25 ps (50 µm) @ -20C
  (35 µm has low bias voltage)

- After 1e15 n/cm^2 :
  35 µm sensor superior to 50 µm sensor due to
  lower rise time and lower bias
  Superior time resolution:
  25 ps (35 µm) vs. 35 ps (50 µm) @ -20C
  21 ps (35 µm) @ -27C
  Reduced bias voltage : 500 V (35 µm) vs. 700 V (50 µm)
  Reduced power for same resolution by large factor (~8x)

First signals out of non-optimized AC-LGAD prototype.

Contributors

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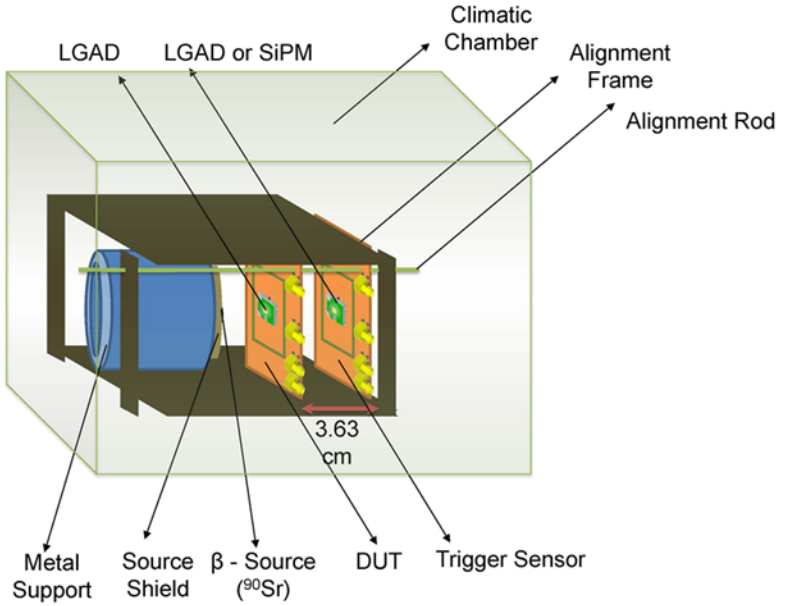
*Students in **bold***

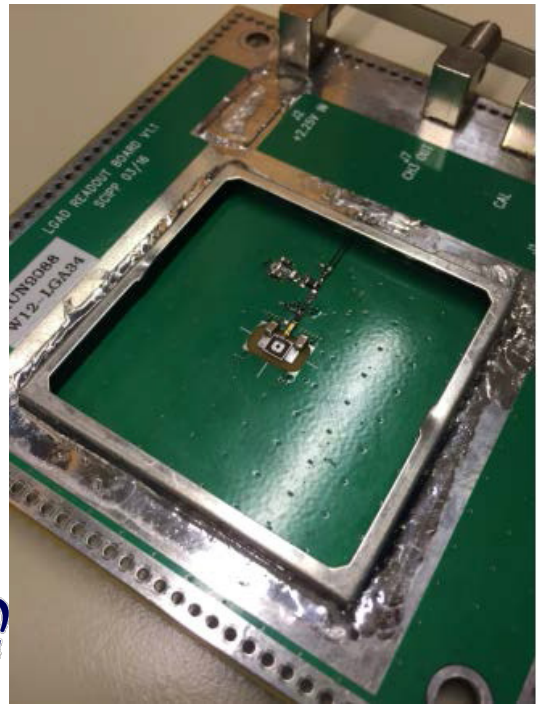
This work was supported by the United States Department of Energy, grant DE-FG02-04ER41286.
Part of this work has been financed by the European Union's Horizon 2020 Research and Innovation funding program, under Grant Agreement no. 654168 (AIDA-2020) and Grant Agreement no. 669529 (ERC UFSD669529), and by the Italian Ministero degli Affari Esteri and INFN Gruppo V.
This work was partially performed within the CERN RD50 collaboration.

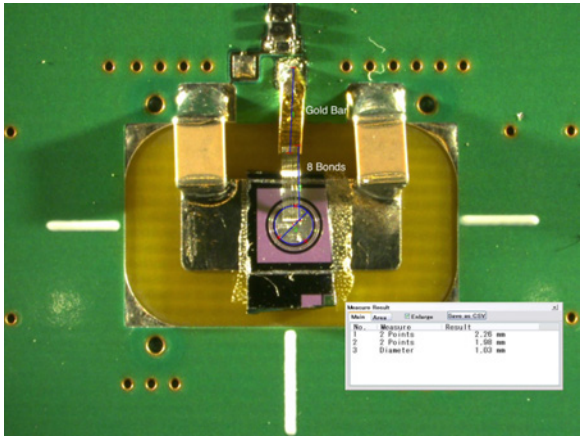
Back-up

UCSC ^{90}Sr β-Source

^{90}Sr β-source Set-up:
DUT LGAD between source and trigger plane
Trigger: either known LGAD
Climate chamber allows operation
between -30C and +20C





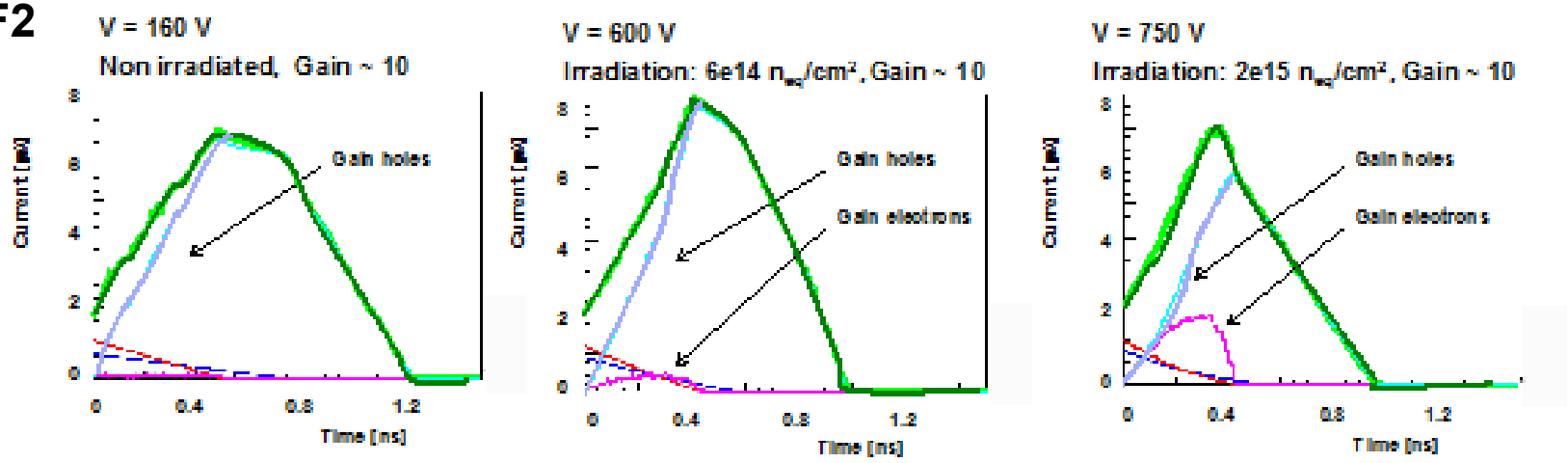


Measurement of pulse shapes with amplifier board
(Low-noise, fast ~3Ghz) developed at UCSC, read Into fast digital scope.

Pulse shapes for irradiated LGAD:

-20C, neutron Irradiation

WF2



Data

